

CN Mid Sem Formula Cheat Sheet

DELAYS

Transmission delay

$$T_{trans} = \frac{L}{R}$$

Propagation delay

$$T_{prop} = \frac{d}{s}$$

Total per hop

$$T = T_{trans} + T_{prop}$$

TRAFFIC INTENSITY

$$I = \lambda \frac{L}{R}$$

Queue delay:

$$T = \frac{L/R}{1 - I}$$

If ($I \rightarrow 1$) $\rightarrow$ delay $\rightarrow \infty$

STORE vs CUT THROUGH

Store $\rightarrow$ full packet $\rightarrow$ safer $\rightarrow$ slower Cut-through $\rightarrow$ partial $\rightarrow$ faster $\rightarrow$ may forward errors

HTTP

Non-persistent:

$$2RTT + T_{trans}$$

Persistent: First object → 2RTT Next objects → 1RTT each

Pipelining → ~1RTT for all objects

HTTP = pull HTTP = stateless HTTP = TCP port 80 HTTP = in-band

SMTP

SMTP = push Port 25 Used to send mail Reading mail → POP3/IMAP

SMTP = in-band

DNS

RR format: (Name, Value, Type, TTL)

A → hostname → IP NS → domain → name server MX → domain → mail server CNAME → alias

DNS uses UDP 53

UDP CHECKSUM

1. Add 16-bit words
2. Wrap carry
3. Invert bits

Receiver → result must be all 1s

Detects 1-bit errors May miss some 2-bit errors

TCP CONGESTION

Slow Start → exponential Congestion Avoidance → +1 MSS per RTT

Timeout:

- $ssthresh = cwnd/2$
- $cwnd = 1$

3 dup ACK:

- $ssthresh = cwnd/2$
- $cwnd = cwnd/2$

Sawtooth graph

WINDOW-LIMITED THROUGHPUT

$$Throughput = \frac{Window}{RTT}$$

If window small → low efficiency

BDP:

$$Bandwidth \times RTT$$

Window must $\geq$ BDP for full utilization

JACOBSON

$$EstimatedRTT = (1 - \alpha)E + \alpha S$$

$$DevRTT = (1 - \beta)D + \beta|S - E|$$

$$Timeout = E + 4D$$

$$\alpha = 0.125 \quad \beta = 0.25$$

GBN vs SR

GBN → retransmit all after loss SR → retransmit only lost

Receiver buffers only in SR

CIRCUIT vs PACKET

Circuit → fixed allocation Packet → statistical multiplexing

Binomial:

$$P(X = k) = \binom{N}{k} p^k (1 - p)^{N-k}$$

TCP FLAGS

SYN → start ACK → acknowledgment FIN → graceful close RST → abrupt close PSH → push URG → urgent pointer valid

LAYERS

Network → routing, IP, logical address Data link → framing, MAC, error detection

Host-to-host → Network Node-to-node → Data link

CONGESTION GRAPH

Load ↑ → Throughput ↑ After saturation → retransmissions ↑ Throughput ↓

Congestion collapse